# 51 VOICE SMART CAR — HARDWARE REVIEW

Scope: Existing PCB is unassembled. This sheet is a review index, not a PCB reconstruction.
No MCU GPIO, power trace, or PCB route is asserted as connected.
Measure the actual board unpowered before wiring or applying power.

## SYSTEM BLOCKS (BOM / photographs)

STC89C52 DIP40 + 11.0592 MHz crystal
Reset and EA networks: PCB connectivity unverified

MCU UART RXD <— one source only (PC/MAX232 OR voice TX)
MCU UART TXD —> MAX232 transmit input (PCB path unverified)

MCU control signals —> L293D inputs/enables (GPIO mapping unknown)
L293D outputs —> M1/M2 (left/right assignment unknown)

DC6V label does not prove a safe MCU logic supply

## DESIGN REVIEW / MEASURE BEFORE USE

[ MCU core ]  socket / crystal / reset / EA; all board routes pending
[ MAX232 ]    RS-232 level converter; verify its 5 V supply rail
[ L293D ]     pins 1/9 enable; pin 16 logic supply; pin 8 motor supply
[ SU—03T ]    candidate offline voice board; variant and UART pending
[ AA pack ]   voltage varies with cell chemistry and discharge state

UART safety: PC/MAX232 and SU—03T must never drive MCU RXD together.
Select modes with power off and measure isolation of the unselected TX.
Initial voice path is one—way TX —> MCU RXD; voltage compatibility pending.

POWER SAFETY: measure DC6V input to IC1—40, IC2—16, IC3—16/8 first.
Confirm regulation, common ground, motor rail, and startup droop before use.

## PIN CHECKLIST (standard IC pins; PCB net assignment unknown)

MCU DIP40: 40 VCC / 20 GND / 18 XTAL2 / 19 XTAL1 / 9 RST / 31 EA
MCU UART: 10 RXD / 11 TXD
L293D: 2,7,10,15 inputs; 1,9 enables; 3,6,11,14 outputs
L293D: 16 VCC1 logic / 8 VCC2 motor / 4,5,12,13 GND
MAX232: 16 VCC / 15 GND / 12,9 R1OUT/R2OUT / 11,10 T1IN/T2IN

Measurement instructions: docs/HARDWARE_AUDIT.md and PINOUT_TRACE.csv.
This sheet has no electrical nets. Empty netlist and clean ERC mean only
that this review file exports; they do not verify PCB wiring, safe power,
or vehicle operation.

Text—only annotated review sheet; not a PCB reconstruction

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**Title: VoiceSmartCar — hardware review sheet**

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